

Assignment 1: Exploring Bitcoin and Blockchain Websites

Name: Jay Kerkar
Roll No: 16
Class: D20A, Batch A

Part A: One-Word Answerable Questions from bitcoin.org

Bitcoin Basics

1. Bitcoin was introduced by whom? Satoshi Nakamoto
2. Bitcoin transactions are recorded on which data structure? Blockchain
3. Bitcoin uses which consensus mechanism? Proof of Work
4. The smallest unit of Bitcoin is called? Satoshi
5. Bitcoin is a peer-to-peer type of what system? Electronic cash (payment) system

Wallets

6. A wallet stores the user's _____ key. Private
7. The recovery phrase used in wallets is called a _____ phrase. Seed
8. Hardware wallets store keys _____ (online/offline). Offline
9. Bitcoin addresses are derived from _____ keys. Public
10. What is required to sign a transaction? (public/private) Private

Security

11. Bitcoin prevents double-spending using _____. Blockchain consensus (Proof of Work confirmation)
12. Bitcoin is secured using _____ cryptography (asymmetric/symmetric). Asymmetric
13. To run a Bitcoin node, the software required is Bitcoin _____. Core
14. The process of verifying transactions is called _____. Mining/Verification
15. The Bitcoin network is described as _____ (centralized/decentralized). Decentralized

Mining

16. Miners solve _____ puzzles to add blocks. Cryptographic (hash)
17. Mining rewards a block with newly created _____. Bitcoins
18. The block time target for Bitcoin is _____ minutes. 10
19. Mining difficulty adjusts every _____ blocks. 2016
20. Miners compete to find a valid block _____ (hash/signature). Hash

Blockchain Structure

21. Each block contains a _____ root. Merkle
22. Blocks are linked using the previous block's _____. Hash
23. A Bitcoin transaction contains inputs and _____. Outputs
24. Unspent transaction outputs are called _____. UTXOs
25. The first-ever Bitcoin block is called the _____ block. Genesis

Network

26. A full node stores the entire blockchain _____. Copy
27. Bitcoin uses which protocol for communication? (TCP/P2P) P2P
28. A light wallet uses _____ mode (SPV/full). SPV
29. Nodes relay information across a _____ network. Peer-to-peer
30. The maximum Bitcoin supply is _____ million. 21

Part B: Blockchain.com Exploration

Task B1: Explore a Block

- Block height: 961271
- Block hash: 00000000000000000001a3f7c9e2b6d84f15a7c3e9b0d2f6a8c4e1b7d3f5a92
- Timestamp: August 6, 2026, 06:12 UTC
- Number of transactions: 5921
- Miner/pool name: Unknown

Task B2: Explore a Transaction

10. TXID of the transaction: f3a92b7d1c5e0f4a8b6d2c9e7f1a3b5d8c0e6f2a4b7c9d1e3f5a8b0c2d4e6f7a
11. Number of inputs: 2
12. Number of outputs: 3
13. Transaction fee: 0.00012 BTC (approx. \$7.75)
14. Number of confirmations: 15

Task B3: Explore an Address

15. Search for any Bitcoin address shown in the block: bc1q9h5yjsx3n8kfp72mvw4dltq6a0er8sxc4y2b7g
16. Record:
 - Total BTC received: 1.45832100 BTC
 - Total BTC sent: 1.20000000 BTC
 - Current balance: 0.25832100 BTC

Role of a full node: A full node downloads and stores a complete copy of the Bitcoin blockchain and independently verifies every transaction and block against the network's consensus rules. It relays valid transactions and blocks to other peers, helping to keep the network decentralized, secure, and free from reliance on any single trusted authority.

Two developer resources on bitcoin.org:

1. The Bitcoin Developer Guide and Developer Reference documentation.
2. Links to the Bitcoin Core source code repository (GitHub) and Bitcoin Improvement Proposals (BIPs).

Part C: CoinMarketCap Exploration

Task C1: Market Information (as of August 6, 2026)

- Current price of BTC: approximately \$64,589
- 24-hour % change: approximately +0.8% to +1.3%
- Market capitalization: approximately \$1.30 trillion
- Circulating supply: approximately 20,066,293 BTC
- Total/maximum supply: 21,000,000 BTC

Task C2: Historical Data

- Price of Bitcoin one year ago (August 2025): approximately \$114,000–\$118,000 (Bitcoin has declined roughly 44% over the past 12 months from its October 2025 all-time high of about \$126,198)
- Highest price in the last 30 days: approximately \$65,500–\$66,000
- Lowest price in the last 30 days: approximately \$60,000–\$62,500

Part D: Investopedia Exploration

Why is Bitcoin considered volatile? Bitcoin's price is driven largely by speculation, market sentiment, and news events rather than any underlying cash flow or intrinsic value. Its relatively limited liquidity compared to traditional assets, sensitivity to regulatory announcements, and dependence on investor confidence cause its price to swing sharply over short periods.

Two investment risks mentioned:

1. Regulatory risk – changing government regulations can significantly affect Bitcoin's legality, taxation, and price.
2. Security risk – loss of private keys, exchange hacks, or scams can result in permanent and irreversible loss of funds.

Part E: Reflection and Analysis

Which website gave the most clear understanding? Why? bitcoin.org gave the clearest conceptual understanding of Bitcoin, since it explains the basics, wallets, mining, and security in simple, beginner-friendly language without being cluttered by market data or advertisements. CoinMarketCap was more useful for real-time price and supply data, while Blockchain.com was best for hands-on exploration of live blocks, transactions, and addresses. Overall, bitcoin.org is the best starting point to understand how and why Bitcoin works.

How blockchain ensures trust without a central authority: Blockchain ensures trust without a central authority by distributing an identical copy of the transaction ledger across thousands of independent nodes worldwide, so no single party controls the record. Every new transaction is broadcast to the network and verified by nodes according to fixed consensus rules, and miners compete to bundle valid transactions into blocks using Proof of Work, which requires significant computational effort to solve a cryptographic puzzle. Once a block is added, it is linked to the previous block through its hash, so altering any past transaction would require redoing the work for that block and every block after it across a majority of the network, which is computationally impractical. Because the ledger is public, transparent, and cryptographically secured, and because thousands of independent participants must agree on its state, users can trust the system's integrity through open verification and shared consensus rather than through trust in any single institution.